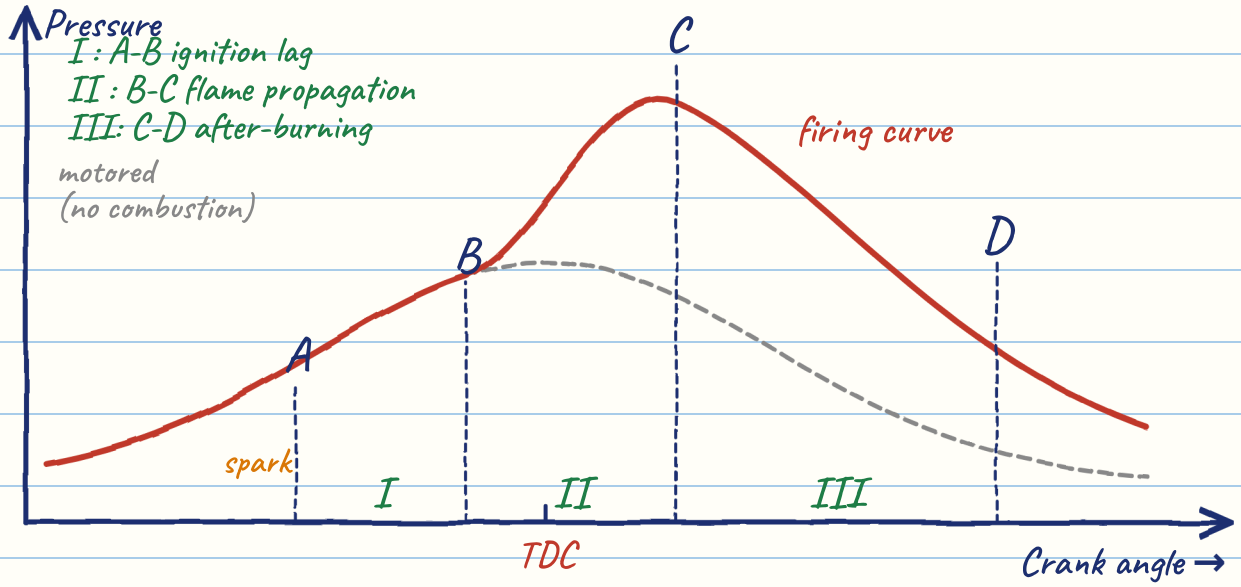
UNIT - III : FUELS AND COMBUSTION

3.1 Combustion in S.I. Engines - Normal Combustion

- In SI engine a **homogeneous** air-fuel mixture is compressed and ignited by a **spark** at ≈ 10-35° before TDC. A flame front starts at the plug & travels through the charge at 10-30 m/s ⇒ turbulent flame propagation.
- Normal combustion = smooth, progressive burning of the whole charge from the spark plug to the far wall without any abnormal event. Pressure rise is ≈ 2-3 bar per degree.



Three stages of combustion

- **Stage I : Ignition lag (A-B)** - spark to first measurable pressure rise; a small nucleus of flame forms (chemical reaction, depends on temperature, pressure, mixture strength, residual gas). ≈ 10% of mass burned. **Pressure not visibly affected.**
- **Stage II : Flame propagation (B-C)** - flame front spreads through the charge; about 90% burned ⇒ rapid pressure rise to the peak, which should occur ≈ 10-15° **after TDC** for maximum power.

- **Stage III : After-burning (C-D)** - burning of the remaining charge in the corners, piston-crown boundary layers, during expansion; pressure falls (little work).

Spark timing is set so that peak pressure occurs at ≈ 10-15° after TDC (MBT - minimum advance for best torque).

3.2 Flame Speed & Importance

- **Flame speed** = rate at which the flame front advances (m/s) through the unburnt mixture. Higher flame speed ⇒ shorter combustion time ⇒ less heat loss, less chance for knock, higher efficiency & permits higher engine speed.
- Flame speed is raised by **turbulence** (swirl, squish, shape of chamber, high rpm) - laminar flame speed of petrol-air ≈ 0.3-0.5 m/s only, turbulent 10-30 m/s.

Effect of engine variables on flame speed / combustion

Variable	Effect
Air-fuel ratio	max. flame speed at slightly rich (≈ 12:1 - 13:1); lean/too rich ⇒ slow
Compression ratio	↑ CR ⇒ ↑ temp. & pressure ⇒ ↑ flame speed, ↑ efficiency
Engine speed	↑ speed ⇒ ↑ turbulence ⇒ flame speed ∝ rpm (time in ms const.)
Spark advance	too early ⇒ negative work, knock; too late ⇒ low peak pressure
Load	↑ load ⇒ less residual gas, ↑ pressure ⇒ ↑ speed, less lag
Chamber shape	compact, central plug ⇒ short flame path, high turbulence

Variable	Effect
Inlet temperature	↑ temp ⇒ ↑ flame speed but ↓ density (↓ power) & ↑ knock
Residual gas	dilutes the charge ⇒ ↓ flame speed

3.3 Abnormal Combustion

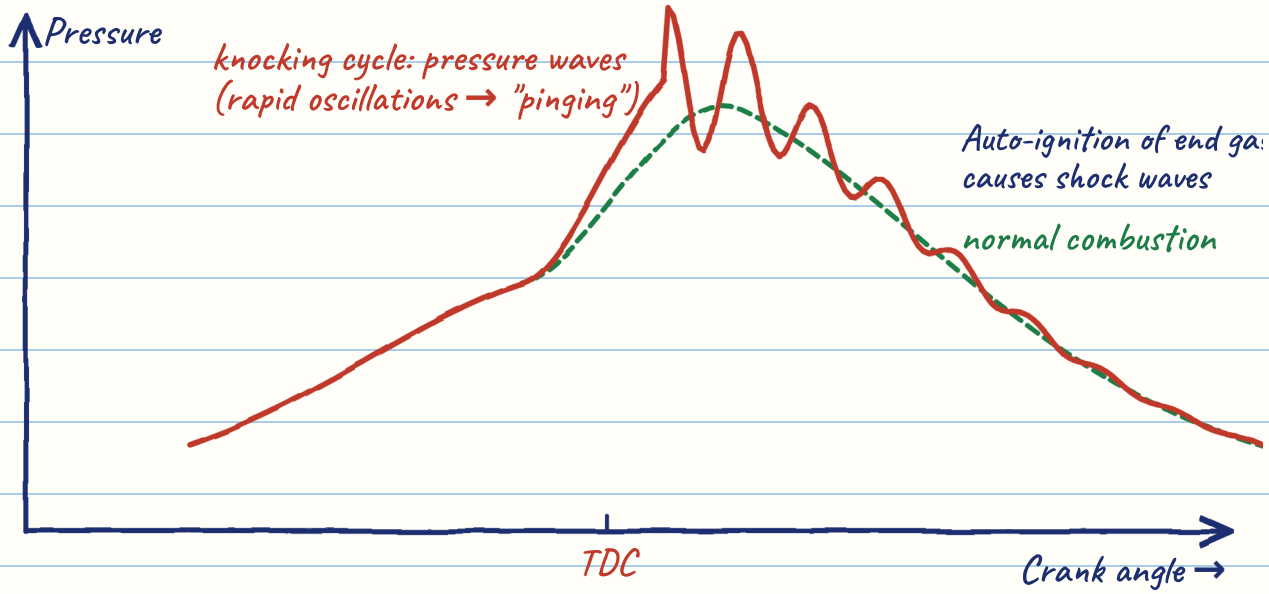
- Any combustion that **departs from the normal pattern**: (a) **Knocking (detonation)**, (b) **Pre-ignition (surface ignition)**, (c) run-on (dieseling), (d) rumble, wild ping, backfire.

(a) Pre-Ignition

- Ignition of the charge **before the spark** occurs, caused by hot spots : overheated spark-plug electrodes, exhaust valve, carbon deposits, sharp edges on the gasket.
- Effects : mixture burns too early ⇒ compression work increases (pressure rise acts against the piston) ⇒ **loss of power**, overheating, a dull thud; can cause piston damage. It can trigger knocking & vice versa (a vicious cycle).
- Remedy : use **correct heat range spark plug**, clean carbon, proper cooling, avoid sharp edges.

(b) Knocking (Detonation) in SI engines

- Normal flame compresses & heats the **end gas** (the last part of unburnt charge) ahead of the flame. If the end gas reaches its **auto-ignition temperature** and the **delay time** is shorter than the time for the flame to arrive, it burns **almost instantaneously** (≈ 1 ms) → violent pressure waves hit the cylinder wall, causing the metallic "ping / knock".



- Effects : **loss of power**, overheating of piston crown & plug, pitting / erosion of piston & head, broken rings, increased heat loss, noise, vibration, higher pollution.

Mechanism : Auto-ignition of the end gas

- Pre-flame reactions form peroxides / aldehydes in the end gas ⇒ chain branching ⇒ spontaneous ignition. High **pressure, temperature & residence time** and low octane fuel (long-chain paraffins) promote knock.

Effect of engine variables on knock

Variable	To reduce knock
Compression ratio	decrease (CR is limited by knock)
Inlet temperature & pressure	decrease (intercooler, no supercharge)
Coolant temperature	decrease (but not too low)
Spark timing	retard the spark
Mixture strength	slightly rich (≈ 10-11:1 best) or lean beyond 18:1
Engine speed	increase (more turbulence, less time)

Variable	To reduce knock
Combustion chamber	compact, central plug, short flame path, cooler end gas
Fuel	high octane number
Engine size	smaller bore (shorter flame travel)
Deposits	remove (reduce effective CR & hot spots)

Pre-ignition vs Knock

Point	Pre-ignition	Knocking
Ignition	before the spark	after spark, in end gas
Cause	hot spot	auto-ignition of end gas
Sound	dull thud	sharp metallic ping
Pressure	rises before TDC	oscillates after peak
Control	clean plugs, cooling	octane, retard, cooler

3.4 Fuel Requirements for SI engine & Fuel Rating

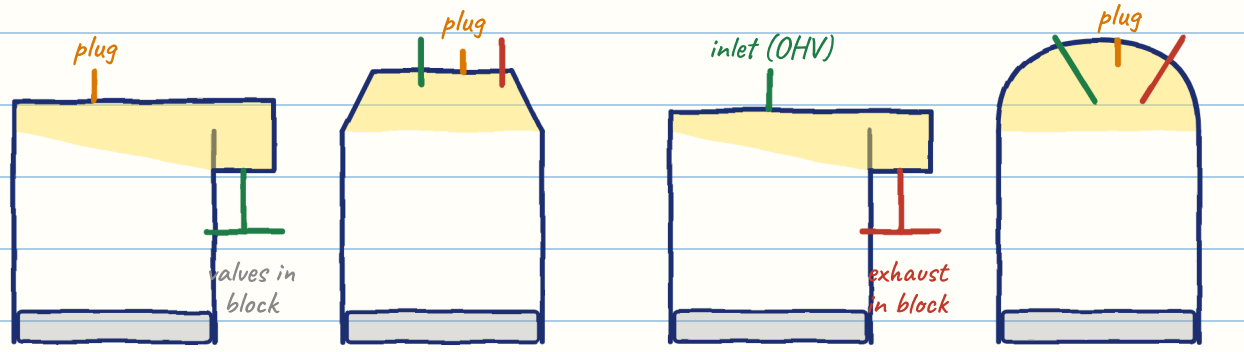
- **Ideal SI fuel** : high **anti-knock** quality (high octane), high volatility (easy starting) but not vapour-lock, high calorific value, good storage stability, low gum, low sulphur, no deposits, cheap.
- **Octane number (ON)** = % by volume of **iso-octane** in a blend of iso-octane (ON = 100) and n-heptane (ON = 0) that matches the knock intensity of the test fuel, in a standard **CFR engine** (variable-compression, single cylinder).
- **RON** (Research, 600 rpm, mild 52 °C) & **MON** (Motor, 900 rpm, 149 °C, severe) ; MON < RON; **Sensitivity** = RON − MON ; Pump rating (R+M)/2 (AKI).
- **Performance number (PN)** for fuels above 100 octane : PN = 100·(knock-limited imep of fuel / of iso-octane).

- **Highest Useful Compression Ratio (HUCR)** : max CR at which fuel runs knock-free.
- Typical: petrol 87-95 RON; LPG, CNG 105-130; ethanol 108.

3.5 Anti-Knock Additives

- **Tetra-ethyl lead (TEL)** : 0.15 - 0.8 ml/l ⇒ stops chain reaction in end gas ; **banned** (pollution, catalyst poisoning); ethylene dibromide scavenger.
- **Tetra-methyl lead**, **MMT** (manganese), **ferrocene** (iron) - metallic.
- **Oxygenates** : **MTBE**, ETBE, ethanol (E10, E20), methanol - raise ON, reduce CO.
- **Aromatics** : benzene, toluene, xylene (limited by emissions); **alkylates & isomerates** (reformed refinery blending).
- Other methods : catalytic reforming, isomerisation, alkylation, polymerisation.

3.6 SI Engine Combustion Chambers



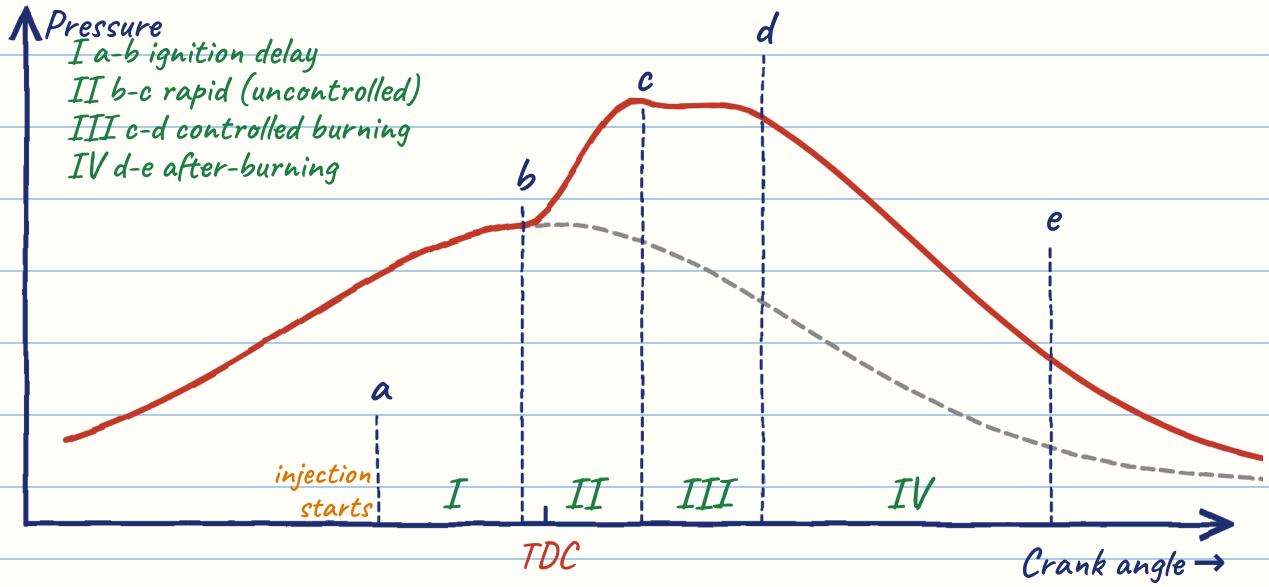
(a) L-head (b) I-head (OHV) (c) F-head (d) Hemispherical

L: simple, but low CR (<7), slow burning · I: compact, higher CR, short flame path
Hemispherical: valves inclined, central plug → shortest flame travel, big valves, best breathing

- **Requirements** : (1) high **power & thermal efficiency**, (2) low knock (compact; short flame travel), (3) high turbulence / swirl, (4) high volumetric efficiency (big valves, good breathing), (5) spark plug located centrally (or near exhaust valve - hottest part), (6) the **end gas** is cooled (large surface/volume ratio near the end), (7) quiet, low deposits, low HC emissions, easy to manufacture.
- **L-head (side valve)** : simple, long flame path; **Ricardo turbulent head** reduces the knock. **I-head / OHV** : valves above; compact, better flow ⇒ high CR. **F-head** : inlet over, exhaust in block (old Rover). **Hemispherical / pent-roof / wedge / bath-tub / Heron** : used with 4-valves per cylinder & central plug (modern).
- **Squish** : the end gas is squeezed into the chamber from the narrow clearance, giving turbulence & cooling the end gas.

3.7 Combustion in C.I. Engines

- **Heterogeneous** combustion : fuel injected (as fine spray) into the hot compressed air (≈ 600-800 °C, 30-40 bar) near the end of compression (≈ 15-25° bTDC); each droplet vaporises, mixes & **self-ignites** ⇒ burning without a spark. Overall A/F lean (20:1 to 70:1).



Four stages of combustion

- I. Ignition delay (a-b) : period between start of injection and start of burning (≈ 0.4-3 ms or 8-15° CA). Has two parts: physical delay (atomisation, heating, vaporisation, mixing) and chemical delay (pre-reactions). Pressure follows the motoring curve.
- II. Rapid (uncontrolled) combustion (b-c) : fuel accumulated during the delay burns almost at once at constant volume ⇒ steep pressure rise (≈ 8-12 bar/deg is the cause of noise).
- III. Controlled (mixing-controlled) combustion (c-d) : the fuel injected burns as it enters; the rate is controlled by the injection rate & mixing ⇒ peak pressure; almost at constant pressure.
- IV. After-burning (d-e) : residual fuel burns during expansion; incomplete burning gives smoke.

3.8 Delay Period – Importance & Effect of Engine Variables

- **Importance** : long delay ⇒ large amount of fuel accumulates ⇒ **very high rate of pressure rise** ⇒ **diesel knock**, rough running, higher peak pressure, stress & noise. Short delay ⇒ smooth, quiet engine.

Variable	Effect on delay period
Compression ratio	↑ CR ⇒ ↑ temp. & pressure ⇒ delay ↓
Intake temp. & pressure	↑ ⇒ delay ↓ (supercharging helps)
Engine speed	delay (ms) ↓ but delay in crank degrees ↑
Injection timing	optimum ≈ 10-15° bTDC; earlier ⇒ longer delay (colder air)
Fuel (cetane no.)	↑ cetane ⇒ delay ↓
Injection pressure, fine atomisation	↓ delay (better mixing)
Load (A/F)	↑ load ⇒ ↑ temp. ⇒ delay ↓
Swirl / turbulence	↑ mixing ⇒ delay ↓

3.9 Diesel Knock

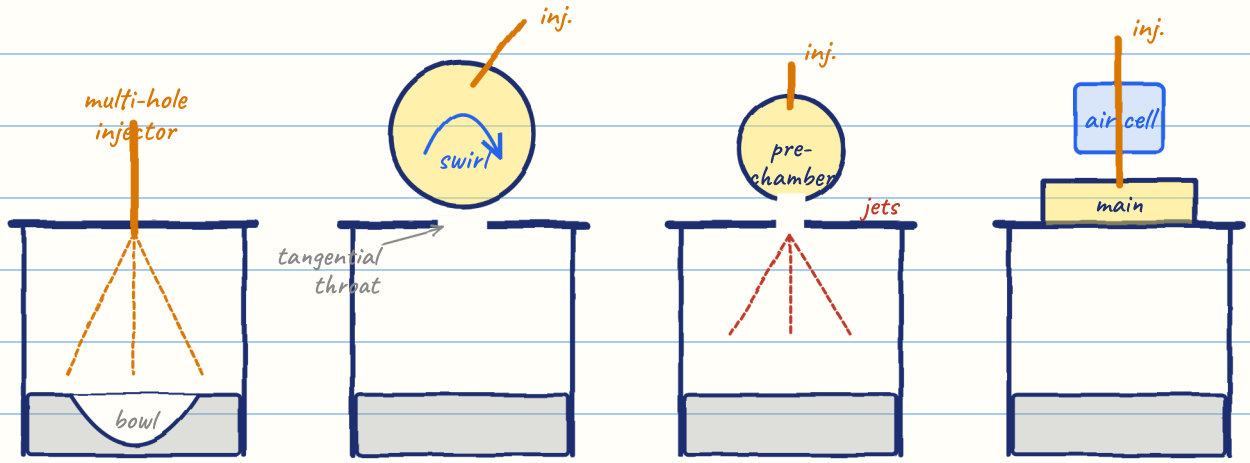
- Diesel knock occurs due to the **sudden ignition of the fuel accumulated during a long delay period** ⇒ abrupt pressure rise (≥ 10 bar/deg) with pressure waves that give the rough knocking sound.
- **Cause** : long delay (cold engine, low cetane, low CR, late injection). It happens at the **start** of combustion (in CI) while in SI knock occurs at the **end** (end gas).

SI knock vs Diesel knock

Point	SI knock	CI (diesel) knock
Where	end of combustion (end gas)	beginning of combustion
Cause	auto-ignition of end gas	too long delay, fuel accumulation
Fuel needed	high octane (high SIT)	high cetane (low SIT)
Remedy	↓ CR, retard, cooling	↑ CR, ↑ intake temp., pilot injection

- **Remedies** : use high-cetane fuel, ↑ compression ratio, ↑ intake temperature/pressure (supercharge), ↑ jacket temperature, **control injection rate** (pilot injection, pintle nozzle), good swirl, correct timing, ignition improvers (amyl-/ethyl-nitrate).

3.10 C.I. Engine Combustion Chambers



(a) Open / DI (b) Swirl chamber (c) Pre-combustion (d) Air cell

DI : high efficiency, hard cold start, noisy · IDI : quiet, smooth, but 10-15 % more fuel

(A) Direct injection (DI) chambers

- Fuel injected directly into the single open chamber in the piston crown; **multi-hole nozzle** (3-10 holes, 200-1000 bar). Types : **open (shallow depth)**, **hemispherical (Hesselman/ Saurer)**, **toroidal / re-entrant (Mexican-hat)** - give swirl by **inlet-port (helical) design**.
- Merits : highest efficiency & lowest fuel consumption, easy cold start (no glow plug), simple head. Demerits : high injection pressure, noisy, sensitive to fuel quality, high NOx.

(B) Indirect injection (IDI) / divided chambers

- Chamber divided into main chamber & a smaller **auxiliary chamber** (≈ 50-80% of clearance volume in swirl chamber); fuel is injected into the auxiliary one, partially burns, then passes through a throat into the main chamber (strong turbulence).
- **Swirl chamber (Ricardo Comet)** : spherical, tangential throat; swirl; speed up to 4500 rpm. **Pre-combustion chamber** : 25-40% volume, restricted passages; quiet. **Air-cell / energy cell** (Lanova) : air stored in the cell enters at the end of burning.
- Merits : low injection pressure (100-150 bar, single-hole pintle nozzle), smooth, quiet, uses wide range of fuels, high speed. Demerits : poor cold start (needs **glow plug**), higher heat loss ⇒ ≈ 10-15% higher fuel consumption.

3.11 Fuel Requirements & Rating of C.I. Fuels

- **Requirements** : high **cetane number** (short delay), good volatility, suitable viscosity (for pump lubrication & spray), high calorific value, low sulphur, low pour / cloud point, good oxidation stability, low carbon residue & ash, cleanliness, non-corrosive.

- **Cetane number (CN)** = % by volume of **cetane (n-hexadecane, CN = 100)** in a mixture with **α-methyl-naphthalene (CN = 0)** that matches the ignition quality of the fuel in a standard CFR engine. (Modern reference: **heptamethylnonane**, CN = 15.)
- **Diesel index / Cetane index** = calculated (API gravity, aniline point) ; typical diesel CN : 45 - 55 (BS : 40 min).
- **Relation** : fuels good for SI have low cetane number and vice versa (high SIT = high octane; low SIT = high cetane).
- **Ignition improvers** : ethyl nitrate, amyl nitrate, di-tertiary butyl peroxide (0.5-1 %).

Summary : SI → octane no., anti-knock, spark ; CI → cetane no., short delay, auto-ignition. Opposite fuel qualities!

3.12 Combustion in SI vs CI Engines

Point	SI engine	CI engine
Mixture	homogeneous	heterogeneous
Start of burning	spark (timed)	auto-ignition (after delay)
Flame	propagates from the plug	many nuclei, spray burning
Load control	quantity of mixture (throttle)	quantity of fuel (A/F varies)
Abnormality	knock (end gas), pre-ignition	diesel knock (long delay)
Fuel quality	octane number	cetane number
Max. CR limit	knock	weight / pressure / stress

Important questions (exam)

- Explain stages of combustion in SI engine with p-θ diagram.
- Differentiate pre-ignition and knocking; explain the effect of engine variables on knocking.

- What is octane number? How is it measured ? Name anti-knock additives.
- Explain the stages of combustion in a CI engine; importance of delay period.
- Explain diesel knock and compare it with SI knock.
- Describe DI and IDI combustion chambers with neat sketches. Define cetane number.